

BI/Data Analyst Portfolio Project

Brand vs Generic Cost Efficiency Analysis

Comparing Prescription Costs and Utilization Across U.S. States

Presented by: Irina Sharapova, Business Intelligence & Data Analyst
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BUSINESS PROBLEM

Prescription drug costs are a major expense for healthcare systems, insurance providers, and patients.

◆ High and Rising Costs

Many medications are available as both expensive brand-name drugs and lower-cost generic alternatives, but prescription and spending patterns can vary between states.

◆ Brand vs Generic Gap

This project analyzes cost and utilization differences between brand and generic medications to better understand potential savings opportunities and prescription trends.

◆ Variation Across States

Prescription utilization and spending patterns vary between states, leading to differences in overall medication costs and generic adoption rates.

Why This Matters

Understanding prescription cost patterns can help identify opportunities for more efficient healthcare spending and improved cost awareness.



Main Questions:

- ➡ Are generic medications consistently cheaper?
- ➡ Which states spend more on brand medications?
- ➡ Which medications show the largest cost differences?
- ➡ Where could healthcare spending potentially be reduced?

UNDERSTANDING BRAND vs GENERIC MEDICATIONS

Brand-name medications are initially protected by patents, but after patent expiration, generic competitors can enter the market.

Generic medications are intended to provide the same therapeutic effect while reducing prescription costs.

Brand Medication



VS

Generic Medication



- Original drug developed by a pharmaceutical company

→ Protected by patent for a period of time

→ Usually more expensive
- Contains the same active ingredient

→ Produced after patent expiration

→ Usually lower cost

Medication Examples Analyzed

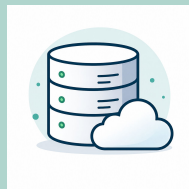
Brand Name	Generic Name	Used For	Why It Matters in This Analysis	U.S. Patent Expiration
Lipitor	Atorvastatin	Lowering cholesterol and reducing risk of heart disease and stroke	One of the most widely prescribed cholesterol medications with a major difference between brand and generic costs	2011
Plavix	Clopidogrel	Preventing blood clots, stroke, and heart attack	Frequently prescribed cardiovascular medication with strong generic utilization trends	2012
Nexium	Esomeprazole	Treating acid reflux and stomach acid conditions	Common long-term medication with noticeable cost differences between brand and generic versions	2014–2015

Dataset & Data Preparation

Medicare Part D Prescriber Public Use File (2014) Accessed through **Google BigQuery Public Datasets**

- Started with **24 million records**
 - and **2,703 medications**

Data Source

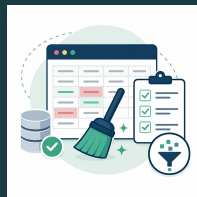


1



Data Cleaning & Preparation

- Selected **3 medication pairs**:
 - **Lipitor ↔ Atorvastatin** (cholesterol)
 - **Plavix ↔ Clopidogrel** (cardiovascular)
 - **Nexium ↔ Esomeprazole** (gastrointestinal)
- Removed non-state location codes (XX, ZZ, GU, AE, AP)
- Classified medications as **Brand, Generic**
- Aggregated by: **State, Drug, Medication type**
- Created Analysis Metrics: **Total claims, Total Drug Cost, Average Cost per Claim, Brand vs Generic Classification**



2



Final Dataset Summary

- Medication Pairs: **3**
- Medications Included: **6**
- 51 Geographic Areas Analyzed (**50 States + DC**)
- Final Analysis Dataset: **287 Rows**

From 24M+ Medicare prescription records to a 287-row analysis-ready dataset

3



Selected three widely prescribed medication pairs representing cholesterol, cardiovascular, and gastrointestinal treatments with available brand and generic alternatives.

Tools Used:

- Google BigQuery,
- SQL,
- Excel,
- Tableau



Project Scope:

- 3 Medication Pairs
- 6 Medications
- 51 Geographic Areas
- Brand vs Generic Comparison

➤ Rows represent aggregated prescription claims grouped by State, Drug, and Medication Type.

➤ Full SQL transformation logic available in Appendix A.

Brand vs Generic Cost Efficiency Analysis

Analysis of Medicare Part D prescription costs across three medication pairs using CMS 2014 data.

Generic medications account for 86.6% of prescriptions and cost approximately 93% less per claim than brand alternatives, representing more than \$1.3 billion in potential Medicare savings.

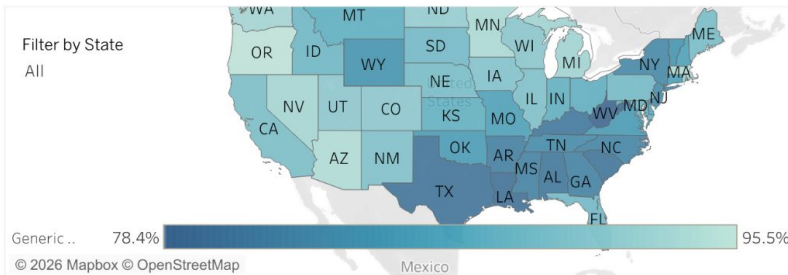


1. Average Cost per Claim: Brand vs Generic Alternatives



2. Generic Usage % by State

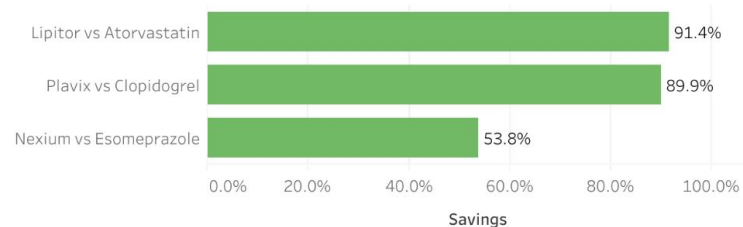
Generic utilization exceeds 78% in every state and reaches over 95% in the highest-adoption states.



3. Potential Savings from Generic Substitution



4. Total Potential Savings %



5. Conclusions & Recommendations:

- Generic prescriptions account for 86.6% of total utilization.
- Generic medications cost approximately 93% less per claim than brand alternatives (\$23 vs \$346).
- Switching Nexium to Esomeprazole generates the largest absolute dollar savings (\$1.3B) due to massive prescription volume, despite having a lower savings percentage (53.8%).
- Lipitor and Plavix show the highest percentage savings opportunities (91.4% and 89.9%).
- Generic utilization exceeds 78% in every state analyzed.

Data Source: CMS Medicare Part D Prescriber Public Use File (2014)
Analysis: Lipitor vs Atorvastatin, Plavix vs Clopidogrel, Nexium vs Esomeprazole

KEY FINDINGS

6



86.6%

Generic
Usage



93%

Lower Cost Per
Claim
(\$23 vs \$346)



\$1.3B

Largest Savings
Opportunity
*Switching Nexium →
Esomeprazole*



78%+

Generic utilization
exceeds 78% in every
state

Key Takeaway: Generic medications deliver significantly lower costs and could generate **more than \$1.3 billion in Medicare savings** through increased substitution.

Business Recommendations:

7

01

Promote generic substitution where clinically appropriate

- Generic drugs already account for 86.6% of utilization
- Significant additional savings remain available

02

Focus on high-volume medications

- Nexium → Esomeprazole shows the largest savings opportunity among the medications analyzed
- High prescription volume creates large cost reduction potential

03

Monitor state-level prescribing patterns

- Generic utilization varies across states
- Best-performing states can provide benchmarks

04

Support physician and patient education

- Increase awareness of lower-cost generic alternatives
- Maintain equivalent therapeutic outcomes

Key Takeaway: Generic alternatives offer substantial cost-saving opportunities while maintaining equivalent therapeutic outcomes.

Thank you

Irina Sharapova
Business Intelligence & Data Analyst

Questions

Final SQL Query(Big Query)

```
SELECT
nppes_provider_state AS state,
drug_name,

CASE
    WHEN drug_name IN ('PLAVIX', 'LIPITOR', 'NEXIUM')
        THEN 'Brand'
    ELSE 'Generic'
END AS medication_type,

SUM(total_claim_count) AS total_claims,

ROUND(
    SUM(total_drug_cost),
    2
) AS total_cost,

ROUND(
    SUM(total_drug_cost) /
    SUM(total_claim_count),
    2
) AS avg_cost_per_claim

FROM `bigquery-public-data.cms_medicare.part_d_prescriber_2014`

WHERE
(
    drug_name = 'PLAVIX'
    OR drug_name = 'CLOPIDOGREL'

    OR drug_name = 'LIPITOR'
    OR drug_name = 'ATORVASTATIN CALCIUM'

    OR drug_name = 'NEXIUM'
    OR drug_name LIKE '%ESOMEPRAZOLE%'
)

AND nppes_provider_state IN (
    'AL', 'AK', 'AZ', 'AR', 'CA', 'CO', 'CT', 'DE', 'FL', 'GA',
    'HI', 'ID', 'IL', 'IN', 'IA', 'KS', 'KY', 'LA', 'ME', 'MD',
    'MA', 'MI', 'MN', 'MS', 'MO', 'MT', 'NE', 'NV', 'NH', 'NJ',
    'NM', 'NY', 'NC', 'ND', 'OH', 'OK', 'OR', 'PA', 'RI', 'SC',
    'SD', 'TN', 'TX', 'UT', 'VT', 'VA', 'WA', 'WV', 'WI', 'WY',
    'DC'
)

GROUP BY
state,
drug_name,
medication_type

ORDER BY
state,
medication_type,
drug_name;
```